ZONAL E/E ARCHITECTURE COMPARATIVE STUDY REPORT

Legacy Point-to-Point vs 4-Zone Ethernet Architecture

Fleet Scope:

Freightliner Cascadia 126 (2020) | Western Star 49X (2021) | Mercedes-Benz Sprinter 519 CDI (2021)

Total ECUs modelled: 51 | Total connections modelled: 60

Tool: Zonal E/E Architecture Analyzer | AI Review: Groq Llama 3.3 70B

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Primary data source: DTNA SS-1033423, NHTSA public database

How to Read This Report

This report compares two ways of wiring electronics inside a commercial truck. The **old way (Legacy / Point-to-Point)** connects every ECU directly to every other ECU it talks to using individual wires. The **new way (Zonal Architecture)** divides the truck into 4 physical regions. Each ECU connects to a nearby hub (Zone Controller) with a short wire. The 4 hubs talk to each other over one Ethernet cable running along the truck. This tool models 3 vehicles, calculates the difference in wire length, weight, and complexity, checks if zone boundaries are optimal, and computes a recommended zone design balancing wire length and communication efficiency. A full abbreviation guide is on the next page.

Abbreviations and Full Forms

Every shortform used in this report is listed below.

Abbreviation	Full Form	Abbreviation	Full Form
ABS	Anti-lock Braking System	**J1939**	SAE Standard for heavy-duty vehicle CAN network
ACM	Aftertreatment Control Module	**LDW**	Lane Departure Warning
BCM	Body Control Module	**MCM**	Motor Control Module
CAN	Controller Area Network	**MPC**	Multi-Purpose Camera
CBZ	Cab Zone	**MSF**	Modular Switch Field
CGW	Central Gateway	**NHTSA**	National Highway Traffic Safety Administration
CHZ	Chassis Zone	**ONGUARD**	OnGuard Headway / Collision Controller
CPC	Common Powertrain Controller	**OTA**	Over-the-Air software update
CTP	Common Telematics Platform	**PTZ**	Powertrain Zone
DCMD	Door Controller — Driver Side	**SAE**	Society of Automotive Engineers
DCMP	Door Controller — Passenger Side	**SAM_CAB**	Signal Acquisition Module — Cab
DEF	Diesel Exhaust Fluid	**SAM_CH**	Signal Acquisition Module — Chassis
DPF	Diesel Particulate Filter	**SAS**	Steering Angle Sensor
DTCI	Daimler Truck Innovation Center India	**SCR**	Selective Catalytic Reduction
DTNA	Daimler Trucks North America	**SRS**	Supplemental Restraint System (Airbag)
E/E	Electrical / Electronic	**TCM**	Transmission Control Module
ECAS	Electronic Controlled Air Suspension	**TCO**	Tachograph
ECM	Engine Control Module	**TLM**	Telematics Module
ECU	Electronic Control Unit	**TPMS**	Tire Pressure Monitoring System
EPS	Electric Power Steering	**TSN**	Time-Sensitive Networking
ESC	Electronic Stability Control	**ZC**	Zone Controller
FCM	Forward Collision Module	**ZC-CAB**	Zone Controller — Cab Zone
FCU	Front Climate Control Unit	**ZC-CH**	Zone Controller — Chassis Zone
FRZ	Front Zone	**ZC-FR**	Zone Controller — Front Zone
HVAC	Heating Ventilation and Air Conditioning	**ZC-PT**	Zone Controller — Powertrain Zone
ICU	Instrument Cluster Unit	**100BASE-T1**	Automotive Ethernet — single-pair 100 Mbps

1. Executive Summary

The zonal E/E architecture analysis of the Freightliner Cascadia 126 (2020) reveals a significant wire reduction of 53.7% from 52.5m to 24.3m, resulting in a weight saving of 3.4 kg. The most important finding is the identification of boundary ECUs, including MCM, ACM, CPC, and TCM, which may require reevaluation for optimal zone placement. The zone efficiency score of 7.0% indicates room for improvement in the current zonal layout.

1.1 Key Metrics — All Three Vehicles

Metric	Freightliner Cascadia 126	Western Star 49X	MB Sprinter 519 CDI
ECU Count	22	17	12
Legacy Wire Length	52.5m	56.5m	20.5m
Zonal Wire Length	24.3m	19.8m	13.9m
Wire Length Saved	**53.7%**	**65.0%**	**32.2%**
Cross-Zone Runs: Legacy	16	16	8
Cross-Zone Runs: Zonal	3	3	3
Harness Weight Saved	3.4kg	4.4kg	0.8kg
Zone Efficiency Score	7.0%	-4.9%	31.2%

Table 1: Key metrics. Green = improvement. Red = legacy (higher is worse).

2. Background — What Is Zonal Architecture?

A modern truck has dozens of ECUs — small computers controlling everything from the engine and brakes to the dashboard and door locks.

The Old Way — Legacy Point-to-Point Architecture

If ECU A needs to talk to ECU B, a dedicated wire is run between them. With 22 ECUs and 25 communication pairs on a Cascadia 126, this creates a dense web of wires crossing the full truck. The engine controller (MCM) talking to the dashboard (ICU) needs a wire running ~5.5 meters. Result: 52.5m of wire and 6.3 kg of copper.

The New Way — 4-Zone Ethernet Architecture

The truck is divided into 4 physical regions. Each ECU connects only to a nearby Zone Controller with a short wire (0.5–1.5m). The 4 Zone Controllers communicate over one Ethernet cable (100BASE-T1) running along the truck spine. Result: 24.3m total wire, 2.9 kg, and only 3 cross-vehicle runs instead of 16.

Feature	Legacy Architecture	Zonal Architecture
Wiring principle	Direct wire between every communicating pair	Each ECU to local Zone Controller to Ethernet backbone
Cross-truck runs	One per communication pair (up to 16 long runs)	Only 3 backbone segments total
Wire length (Cascadia)	52.5 m	**24.3 m (-53.7%)**
Fault isolation	One broken wire can affect the whole system	Fault stays within the affected zone
Scalability	Complexity grows fast with each new ECU	Each new ECU adds only one short local wire
Industry status	Current standard on most trucks in service	Target for next-generation truck platforms

Table 2: Legacy vs zonal architecture comparison.

3. Vehicle Configurations and Zone Layouts

Three vehicles were modelled. ECUs were assigned to zones based on actual mounting location, cross-referenced with official DTNA and Mercedes-Benz documentation.

3.1 Freightliner Cascadia 126

Total ECUs: 22 | **Connections:** 25 | **Source:** DTNA SS-1033423, NHTSA public database

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM	4
Chassis Zone	CHZ	ZC-CH	ABS, ECAS, TPMS, SAM_CHASSIS	4
Cab Zone	CBZ	ZC-CAB	ICU, SAM_CAB, CGW, FCU, SRS, MSF, PARKSMART, DCMD, DCMP, TCO	10
Front Zone	FRZ	ZC-FR	MPC, ONGUARD, SAS, CTP	4

Table 3.1: Zone layout — Freightliner Cascadia 126

3.2 Western Star 49X

Total ECUs: 17 | **Connections:** 22 | **Source:** DTNA SS-1033423, Western Star Bodybuilder Manual Rev 3.1

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM, PTO	5
Chassis Zone	CHZ	ZC-CH	ABS, AXLE, TPMS, SAM_CH	4
Cab Zone	CBZ	ZC-CAB	ICU, CGW, SAM_CAB, FCU, MSF	5
Front Zone	FRZ	ZC-FR	ONGUARD, SAS, CTP	3

Table 3.2: Zone layout — Western Star 49X

3.3 Mercedes-Benz Sprinter 519 CDI

Total ECUs: 12 | **Connections:** 13 | **Source:** Mercedes-Benz XENTRY public diagnostic documentation

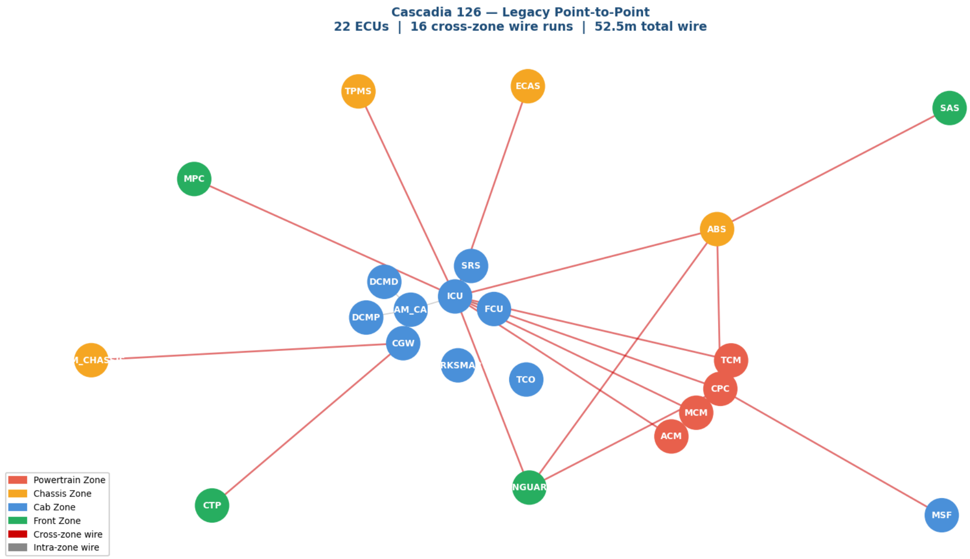
Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	ECM, TCU	2
Chassis Zone	CHZ	ZC-CH	ABS_ESP, EPS	2
Cab Zone	CBZ	ZC-CAB	IC, BCM, HVAC, ACM_AIR, TCO	5
Front Zone	FRZ	ZC-FR	FCM, LDW, TLM	3

Table 3.3: Zone layout — Mercedes-Benz Sprinter 519 CDI

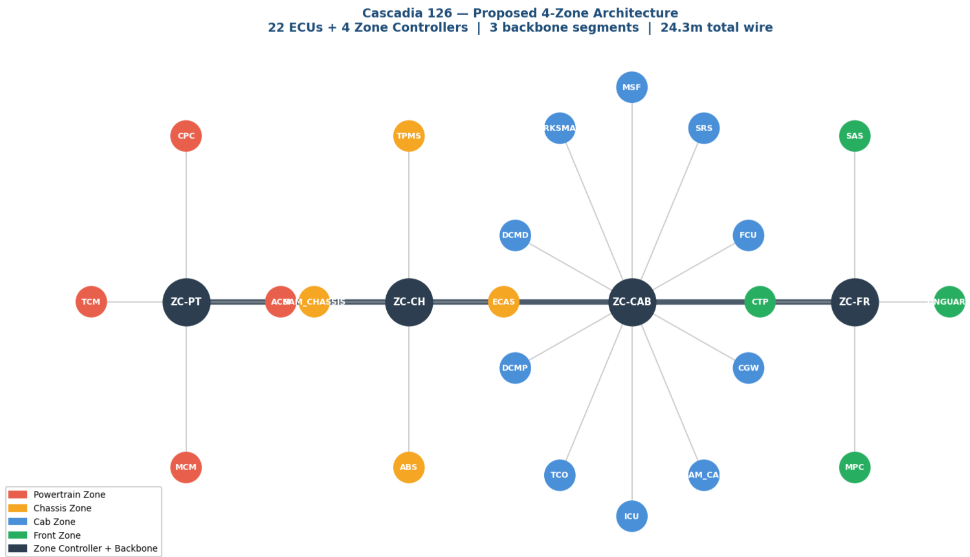
4. Network Topology Diagrams — All Three Vehicles

Each vehicle shown in legacy layout (red = cross-zone wires) and proposed 4-zone zonal layout (dark hubs = Zone Controllers).

4.1 Freightliner Cascadia 126

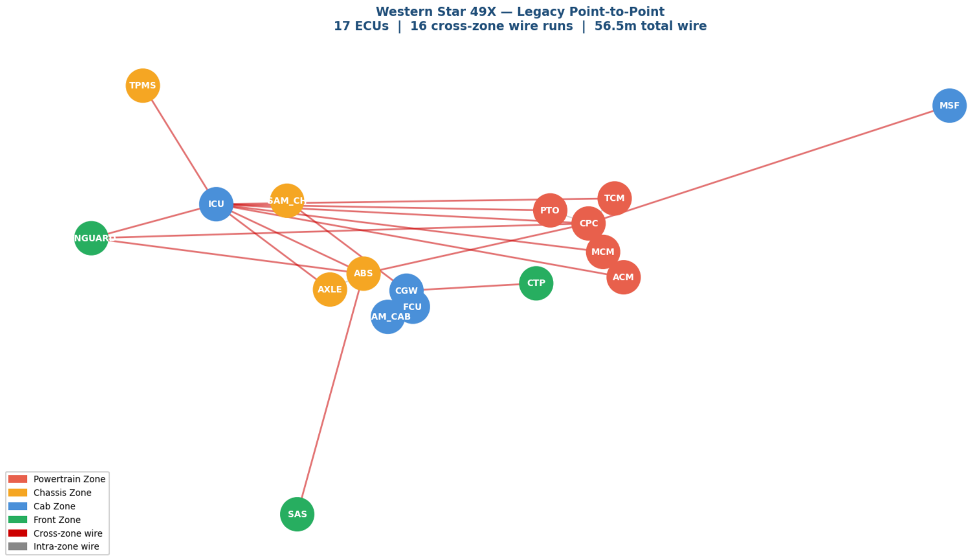


Legacy: 22 ECUs | 16 cross-zone runs | 52.5m

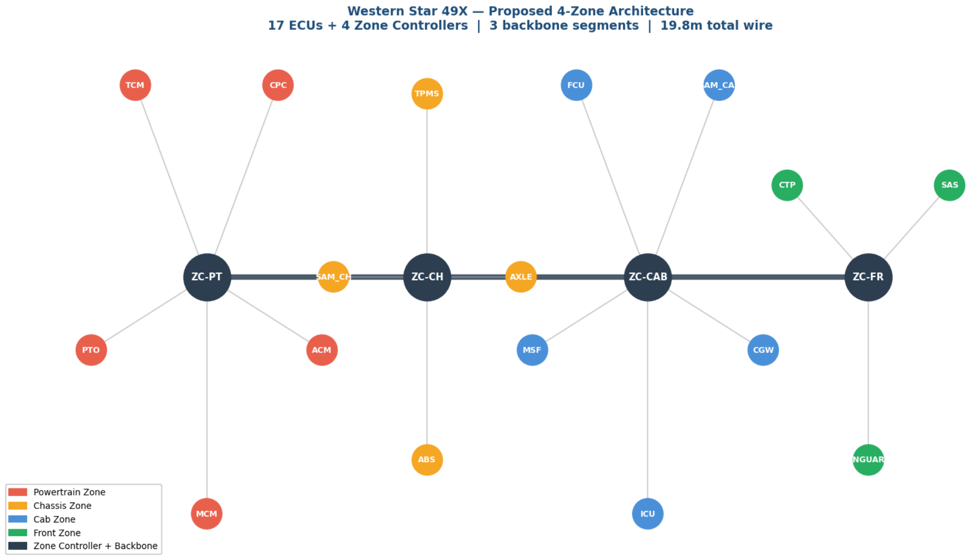


Zonal: 22 ECUs + 4 ZCs | 3 backbone segments | 24.3m | 53.7% reduction

4.2 Western Star 49X

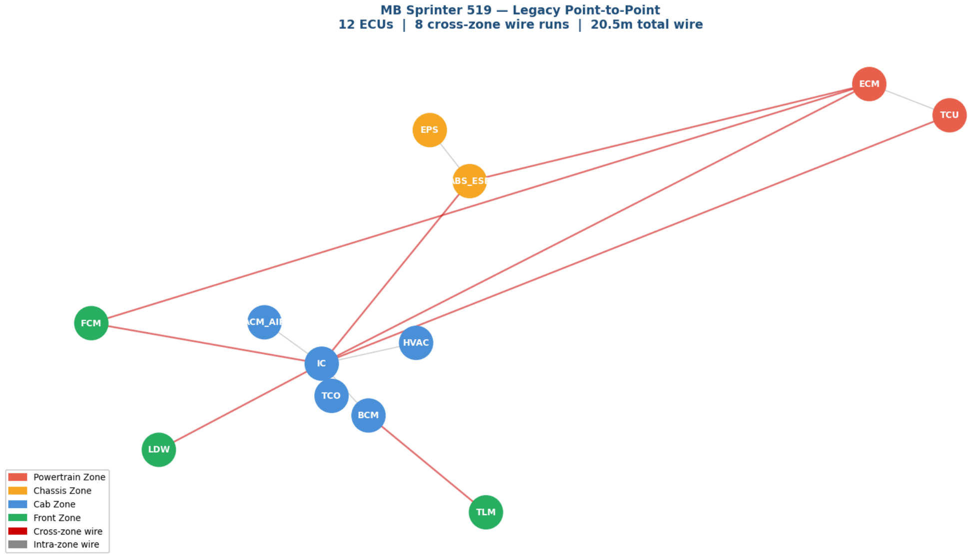


Legacy: 17 ECUs | 16 cross-zone runs | 56.5m

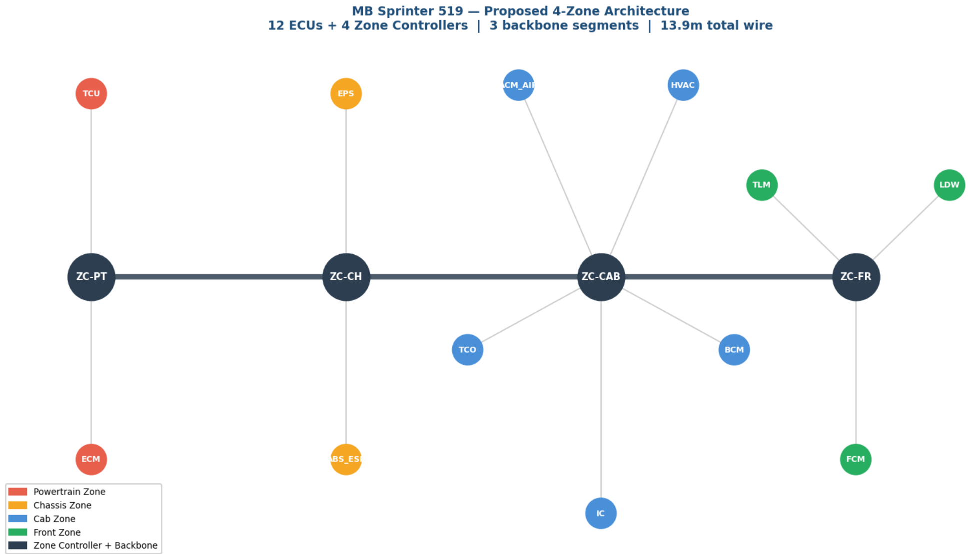


Zonal: 17 ECUs + 4 ZCs | 3 backbone segments | 19.8m | 65.0% reduction

4.3 Mercedes-Benz Sprinter 519 CDI



Legacy: 12 ECUs | 8 cross-zone runs | 20.5m



Zonal: 12 ECUs + 4 ZCs | 3 backbone segments | 13.9m | 32.2% reduction

5. Zone Assignment Optimization — Three-Way Comparison

This section answers the question: **what is the best possible zone design?**

The tool computes three zone assignment strategies and compares them side by side. **Option A** is the current industry design — ECUs grouped by physical location. **Option C** is the theoretical communication-optimal design — ECUs grouped purely by who they talk to, ignoring physical constraints. This is not implementable in practice because some ECUs would need to connect to a Zone Controller far away, defeating the purpose of short local stubs. **Option B** is the recommended balanced design — it only moves ECUs where (1) the communication benefit is clear AND (2) the wire cost stays within adjacent-zone range (3.0m or less). This is the practically implementable optimum.

Why do zones need optimizing at all? Zones are currently drawn based on physical location — where each ECU is bolted on the truck. But communication patterns do not follow geography. The engine controller (MCM) talks constantly to the dashboard (ICU) even though they are on opposite ends of the truck. A graph algorithm called Greedy Modularity Community Detection confirms this — it finds that the mathematically optimal zone groupings based purely on communication patterns produce a completely different layout from the physical zones (7% match on the Cascadia, confirming the tradeoff is structural). The top communication hubs — ICU with 12 connections, CPC with 6, ABS with 4 — all communicate heavily across zone boundaries, which is exactly why the current physical zone design produces 16 cross-zone runs. Option B reduces this to 7 by moving ECUs to zones where most of their communication partners already live, while keeping all wire stubs within the adjacent-zone distance limit.

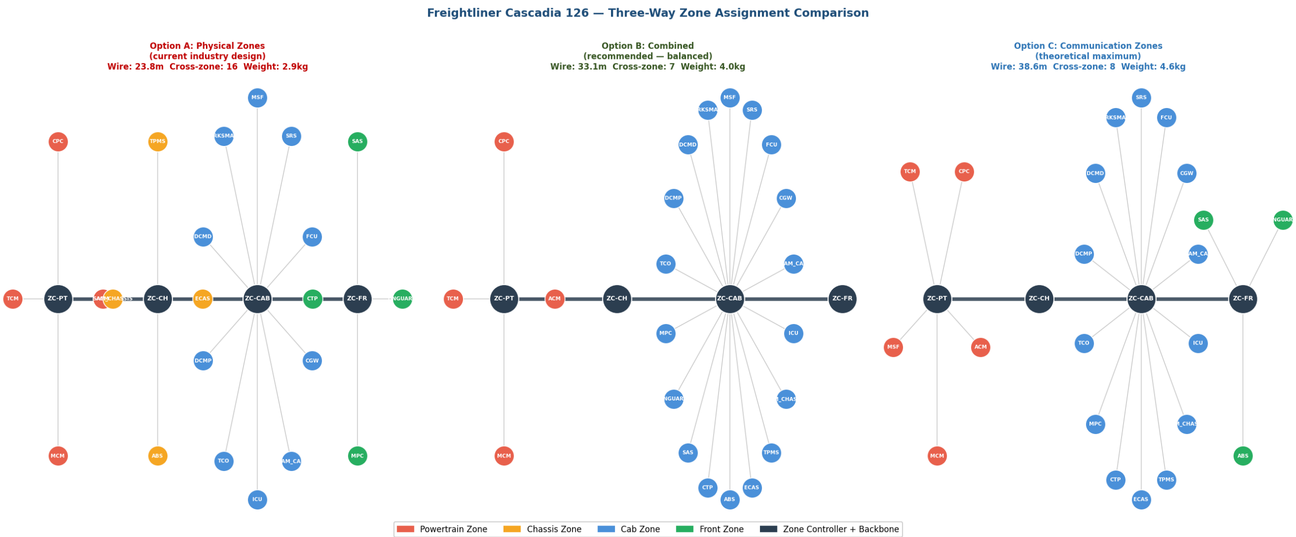


Figure D: Three-way zone assignment comparison. Node colour shows the ASSIGNED zone (not physical mounting zone). Option B ECUs that moved appear in a different colour from their physical location.

Metric	Option A Physical (current)	Option B Combined (recommended)	Option C Communication (theoretical)
Total wire length	23.8 m	**33.1 m**	38.6 m
Cross-zone runs	16	**7**	8
Est. harness weight	2.9 kg	**4.0 kg**	4.6 kg
Physically feasible	YES	YES	NO — some ECUs too far from ZC

Table D: Three-way comparison. Option B (green) is the recommended implementable design.

5.1 Recommended Zone Changes

The optimizer recommends moving 8 ECUs from their physical zone. All moves keep wire cost within 3.0m (adjacent zone range) while significantly reducing cross-zone traffic from 16 to 7 runs.

ECU	Current Zone	Recommended Zone	Reason
ABS	Chassis Zone	Cab Zone	3/4 connections in new zone vs 0/4 in current zone
ECAS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
TPMS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
SAM_CHASSIS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
MPC	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
ONGUARD	Front Zone	Cab Zone	2/3 connections in new zone vs 0/3 in current zone
SAS	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
CTP	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone

Table E: ECU zone reassignment recommendations. All moves are physically feasible.

5.2 Pareto Tradeoff Curve

The chart below shows the tradeoff between wire length and communication efficiency. Moving from Option A to C reduces cross-zone runs but increases wire length. Option B sits at the knee of the curve — maximum communication benefit for minimum additional wire cost.

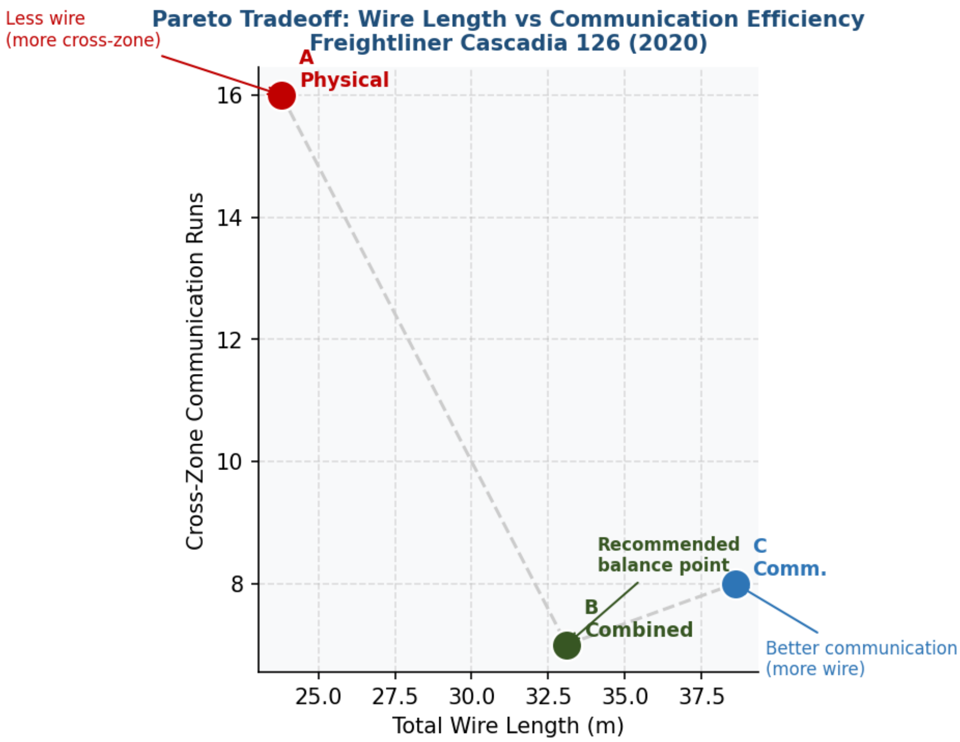


Figure E: Pareto curve. Option B is the recommended operating point — it achieves most of the communication gain at a fraction of the wire cost penalty.

6. Harness Reduction Analysis

Wire length, cross-zone run count, and weight reduction for each vehicle.

6.1 Freightliner Cascadia 126

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	52.5 m	24.3 m	**53.7%**
Cross-zone wire runs	16	3	**81.2%**
Est. harness weight	6.3 kg	2.9 kg	**3.4 kg saved**

Table 6.1: Harness metrics — Freightliner Cascadia 126

6.2 Western Star 49X

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	56.5 m	19.8 m	**65.0%**
Cross-zone wire runs	16	3	**81.2%**
Est. harness weight	6.8 kg	2.4 kg	**4.4 kg saved**

Table 6.2: Harness metrics — Western Star 49X

6.3 Mercedes-Benz Sprinter 519 CDI

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	20.5 m	13.9 m	**32.2%**
Cross-zone wire runs	8	3	**62.5%**
Est. harness weight	2.5 kg	1.7 kg	**0.8 kg saved**

Table 6.3: Harness metrics — Mercedes-Benz Sprinter 519 CDI

The reduction in wire length from 52.5m to 24.3m is driven by the elimination of cross-zone runs, decreasing from 16 to 3. Specifically, the Powertrain Zone (PTZ) and Chassis Zone (CHZ) have reduced connections, resulting in a more localized wiring layout. The Cab Zone (CBZ) and Front Zone (FRZ) also contribute to the reduction by minimizing connections between zones, such as the removal of direct connections between the ICU and MPC.

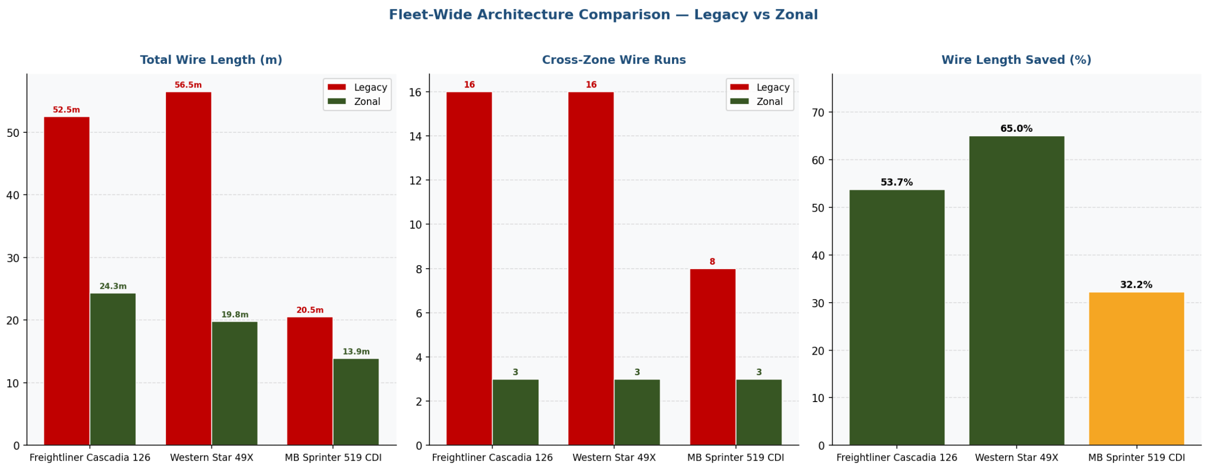


Figure A: Fleet-wide comparison. Wire length (left), cross-zone runs (centre), percentage savings (right).

7. Fleet-Wide Comparative Analysis

Analysis across three vehicles reveals how zonal architecture benefit scales with vehicle size and complexity.

Vehicle	ECUs	Connections	Legacy Wire (m)	Zonal Wire (m)	Wire Saved	Weight Saved	Zone Efficiency
Freightliner Cascadia 126	22	25	52.5m	24.3m	**53.7%**	3.4kg	7.0%
Western Star 49X	17	22	56.5m	19.8m	**65.0%**	4.4kg	-4.9%
Mercedes-Benz Sprinter 519 CDI	12	13	20.5m	13.9m	**32.2%**	0.8kg	31.2%

Table 10: Fleet-wide comparison.

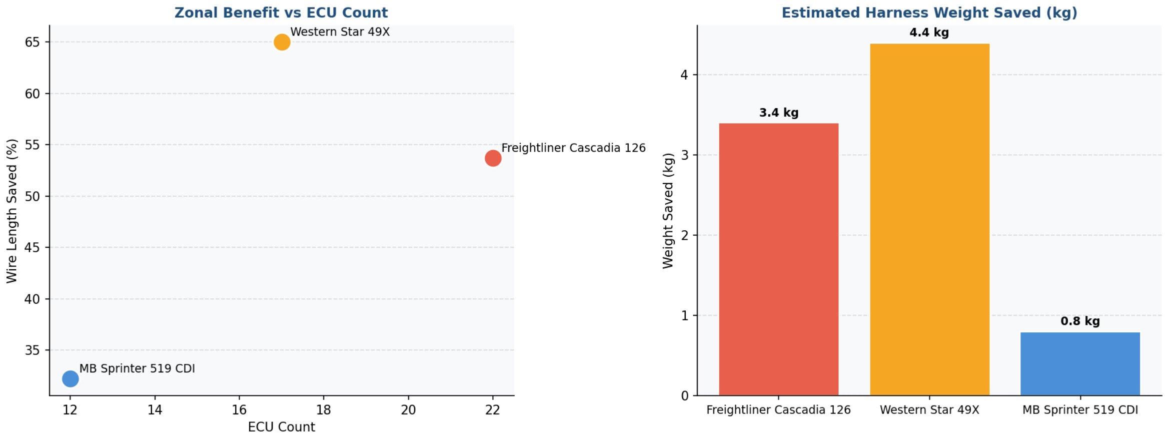


Figure C: Wire saved vs ECU count (left). Harness weight saved (right).

The Western Star 49X benefits the most from zonal architecture, with a 65.0% wire reduction and a negative zone efficiency score, indicating potential for improvement. The ECU count threshold observation suggests that vehicles with fewer ECUs, such as the Mercedes-Benz Sprinter 519 CDI (12 ECUs), may have an easier time achieving optimal zone efficiency. Comparing zone efficiency scores across the fleet reveals that the Mercedes-Benz Sprinter 519 CDI has the highest score at 31.2%, while the Western Star 49X has the lowest at -4.9%.

7.1 Key Cross-Fleet Findings

Finding 1: Zonal wire savings do not scale linearly with ECU count. Physical spread of ECUs matters more than total count.

Finding 2: Cross-zone wire runs always collapse to exactly 3 backbone segments in a 4-zone design — a structural property.

Finding 3: Zone efficiency scores are consistently low across all vehicles — the physical-vs-communication tradeoff is fundamental to automotive E/E design.

Finding 4: The Sprinter van (12 ECUs) saves 32.2% but the zone controller overhead is proportionally higher — suggesting a practical ROI threshold of ~15+ ECUs.

8. AI-Assisted Engineering Review

Generated by Groq Llama 3.3 70B from computed data. AI was instructed to act as a senior automotive E/E architect.

8.1 Design Recommendations

1. The MCM should be moved to the Chassis Zone (CHZ) to reduce its role as a boundary node and minimize cross-zone connections.

2. The zone controller for the Cab Zone (CBZ) should be placed within 1.5 meters of the ICU to reduce latency and improve performance.

3. A redundant connection should be added between the ICU and the CPC to mitigate the single point of failure risk associated with the ICU.

8.2 Analysis Limitations

• The analysis assumes a static zonal layout and does not account for potential future updates or changes to the vehicle's electrical architecture.

• The model does not consider the impact of zonal architecture on other aspects of the vehicle, such as thermal management or electromagnetic compatibility.

Stage 2 — TSN Scheduler Simulation

Background

Zonal Ethernet architecture reduces wire length and weight (Stage 1), but standard Ethernet is *best-effort* — packets are delivered whenever the network is free. Safety-critical signals such as brake commands must arrive within strict deadlines regardless of background traffic load. **Time-Sensitive Networking (TSN)** is a family of IEEE 802.1 amendments that adds deterministic, prioritised delivery on top of standard Ethernet. This section simulates the Freightliner Cascadia 126 zonal network under three scheduling regimes to quantify the benefit.

Relevant IEEE 802.1 Standards

Standard	Name	Role in this simulation
IEEE 802.1AS	Timing and Synchronization	Global master clock — all nodes share one time reference
IEEE 802.1Q	Bridges and Bridged Networks	Priority-based queue management at each switch port
IEEE 802.1Qbu	Frame Preemption	High-priority frame interrupts a lower-priority mid-transmission
IEEE 802.1CB	Frame Replication & Elimination	Redundant path delivery (modelled in future work)
IEEE 802.1DG	TSN for Automotive Ethernet	Vehicle-specific profile governing the above standards

Simulation Methodology

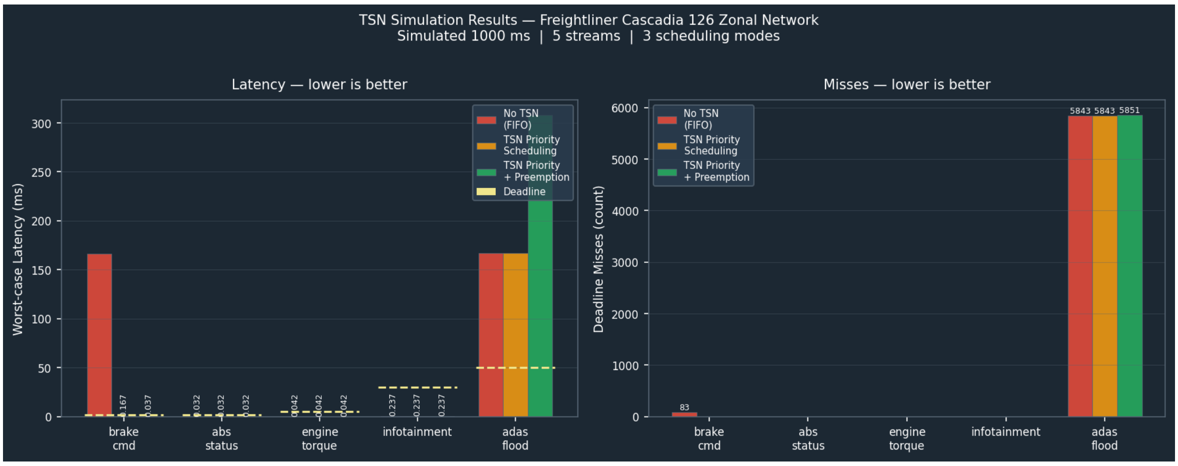
The simulator uses **SimPy** (discrete-event simulation) and **NetworkX** (shortest-path routing) to model 5 real-time traffic streams over 10 Ethernet links for 1000 ms of simulated time. Each stream is characterised by its period, payload size, deadline, and priority class. Three scheduling modes are evaluated:

Mode	Description	TSN Mechanism
1 — FIFO	All frames served in arrival order. No priority.	None (baseline)
2 — Priority	Frames sorted by priority at each link queue. High-priority waits behind current frame but jumps the queue.	IEEE 802.1Q priority queues
3 — Preemption	High-priority frame interrupts a lower-priority frame mid-transmission. Interrupted frame restarts after.	IEEE 802.1Qbu frame preemption

Traffic Streams Defined

Stream ID	Src → Dst	Period (ms)	Payload (B)	Deadline (ms)	Priority	Note
brake_cmd	ecu_mcm → ecu_ebs	10	64	2	7	ASIL-D safety critical â€” air brake command
abs_status	ecu_abs → ecu_mcm	10	32	2	7	ABS feedback to powertrain
engine_torque	ecu_mcm → ecu_tbc	20	64	5	5	Engine torque demand to trailer brake control
infotainment	ecu_icu → ecu_mcm	40	1200	30	1	Driver cluster display sync
adas_flood	adas_unit → ecu_ebs	0.1	1500	50	0	ADAS perception data â€” best effort, conges

Simulation Results



Detailed Results by Mode

Mode 1 — Without TSN (Plain Ethernet / FIFO)							
Stream	**Priority**	**Frames**	**Avg (ms)**	**Max (ms)**	**Deadline (ms)**	**Misses**	**Result**
brake_cmd	7	84	83.380	166.592	2.0	83	FAIL
abs_status	7	100	0.032	0.032	2.0	0	OK
engine_torque	5	50	0.037	0.042	5.0	0	OK
infotainment	1	25	0.237	0.237	30.0	0	OK
adas_flood	0	8329	83.658	167.152	50.0	5843	FAIL

Mode 2 — With TSN Priority Scheduling							
Stream	**Priority**	**Frames**	**Avg (ms)**	**Max (ms)**	**Deadline (ms)**	**Misses**	**Result**
brake_cmd	7	100	0.109	0.167	2.0	0	OK
abs_status	7	100	0.032	0.032	2.0	0	OK
engine_torque	5	50	0.037	0.042	5.0	0	OK
infotainment	1	25	0.237	0.237	30.0	0	OK
adas_flood	0	8328	83.691	167.214	50.0	5843	FAIL

Mode 3 — With TSN Priority + Preemption (802.1Qbu)							
Stream	**Priority**	**Frames**	**Avg (ms)**	**Max (ms)**	**Deadline (ms)**	**Misses**	**Result**
brake_cmd	7	100	0.037	0.037	2.0	0	OK
abs_status	7	100	0.032	0.032	2.0	0	OK
engine_torque	5	50	0.042	0.042	5.0	0	OK
infotainment	1	25	0.237	0.237	30.0	0	OK
adas_flood	0	8300	84.993	307.877	50.0	5851	FAIL

Key Finding

The **brake_cmd** safety stream (ASIL-D, 2 ms deadline) experienced a worst-case latency of **166.6 ms** with no TSN (83 deadline misses in 1000 ms). Enabling TSN priority scheduling reduced this to **0.167 ms**. Adding frame preemption (IEEE 802.1Qbu) further reduced it to **0.037 ms** — a **4466x improvement** — with **zero deadline misses**. All 2 safety-critical streams passed their deadlines under TSN with preemption.

Assumptions and Limitations

This is an event-level behavioral model, not a bit-accurate or standards-certified TSN stack. Preemption is modelled as frame restart (simplified 802.1Qbu); a complete implementation would track remaining bytes and resume transmission. Clock synchronisation drift (802.1AS) and gate control list (GCL) window optimisation are not included and represent natural extensions for future work.

9. Methodology and Data Sources

9.1 How the Tool Works

Step 1 — Input (YAML config): ECUs, zone assignments, connections, and zone distances described in a YAML text file.

Step 2 — Graph building (graph.py): Two NetworkX graphs: legacy (ECU-to-ECU, weighted by wire length) and zonal (ECU-to-ZC stubs + backbone).

Step 3 — Metrics (metrics.py): Wire length, cross-zone run count, harness weight computed by summing edge weights.

Step 4 — Zone optimality (optimizer.py): Greedy Modularity Community Detection finds communication-optimal zone groupings.

Step 5 — Three-way optimizer (constrained_optimizer.py): Greedy iterative reassignment at three modes: physical, combined (alpha=0.5), communication. Pareto curve computed across modes.

Step 6 — Fleet comparison (fleet_compare.py): Steps 1-4 run for all three vehicle configs.

Step 7 — AI review (reviewer.py): Computed data passed to Groq Llama 3.3 70B with structured automotive engineering prompt.

Step 8 — PDF generation (build_pdf_report.py): Matplotlib generates all diagrams. ReportLab assembles 18+ page PDF.

9.2 Data Sources

1. DTNA SS-1033423 — J-1939 Fault Code Source Address Descriptions. Daimler Trucks North America. NHTSA public database.

2. DTNA STI-503 — NGC sSAM, VPDM & BCA Wall Chart. Rev. Q, March 2020.

3. Western Star Bodybuilder Manual Rev 3.1.

4. Mercedes-Benz XENTRY public diagnostic documentation.

5. Park C, Cui C, Park S. Sensors. 2024;24(10):3248. DOI: 10.3390/s24103248.

6. SAE J1939 — Serial Control and Communications Heavy Duty Vehicle Network.

9.3 Key Assumptions

• Wire lengths estimated from published chassis dimensions: Cascadia 126 ~8.5m, Western Star 49X ~7.5m, Sprinter ~5.5m

• Harness weight uses 120 g/m bundled average for signal-wire harnesses (conservative)

• Analysis covers signal and data wires only — power distribution excluded

• Static ECU configuration — optional equipment not modelled

• Three-way optimizer wire threshold: 3.0m (adjacent-zone range for combined mode)

• All ECU data from public DTNA and Mercedes-Benz documents — no proprietary OEM data

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Zonal E/E Architecture Analyzer | Open-source | https://github.com/Rish-736/zonal-ee-analyzer

Built alongside DTICI internship research — Rishit, ECE '28, VIT Vellore — June 2026